

M002 Week 1 — English Worksheet

Topic: While Loops — Move Until You Hit a Wall · **Course:** Maze Madness · **Time:** about 45 minutes

This week your agent uses a **while loop** to keep moving forward until it hits a wall.

1 · Predict 🧠

Read each set of steps. Before you imagine the agent doing it, write what you think will happen.

```
while no wall ahead:
    move forward
```

When does the agent stop? Why?


```
move forward
move forward
move forward
```

How is this different from the while loop above? What if the wall is only 2 steps away?


```
while no wall ahead:
    move forward
turn right
while no wall ahead:
    move forward
```

The agent finishes the first while loop. What happens next?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The agent is supposed to keep moving forward **until** it hits a wall.

```
while wall ahead:
    move forward
```

Hint: the condition is backwards.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The agent is supposed to move forward step by step. Right now it only checks the wall **once**, before moving 10 steps no matter what.

[illegible]

Write the fixed code using a while loop:

Why was it wrong? Why does your while loop work better?

Bug C — The agent is supposed to stop **right in front** of the wall, not crash into it.

```
while no wall ahead:  
    move forward  
    move forward
```

Hint: the agent moves twice before checking again.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Use a **while loop** to move the agent from the start of the straight maze all the way to the wall. When you finish, come back here.

Send a photo or video of the agent reaching the wall, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I read the maze and noticed ...

My while loop checked ...

The agent stopped because ...

The hardest part was ...

To fix it, I ...

A while loop is useful because ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while the agent runs the maze. Talk through it like you are teaching someone who has never seen it. Try to use these words: **while**, **loop**, **detect**, **wall**, **forward**.

1. Show the start of the maze, then run your code.
2. Read your while loop out loud and say what makes it stop.
3. Show one bug you hit and how you fixed it.
4. Say why a while loop is better than writing `move forward` many times.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.